

Project 1: Database Planning and Requirement Analysis

Due date: 28 January 2026 before 9 p.m.

This project requires students to work as a team of no more than three to design, implementation and document a relational database system for a real-world scenario. The objective is to apply theoretical concepts to a practical problem. The project is divided into five major milestones: (1) Database Planning and Requirement Analysis, (2) Conceptual Database Design, (3) Logical Database Design, (4) Implementing Database and (5) Physical Database Design and Tuning.

Objective:

In the first milestone, the project aims to establish a clear understanding of the project's domain, define the scope, and gather comprehensive requirements that will guide the conceptual and logical design of the database application.

Description:

In this initial phase, students will engage in a structured process to analyze the problem domain and identify the data needs of the target application. The goal is to ensure that the database application is purpose-built, scalable, and aligned with user expectations and business rules. This phase includes: (1) Identifying stakeholders and user roles, (2) understanding workflows and data interactions, (3) defining functional and non-functional requirements, (4) outlining constraints, assumptions, and success criteria

Key Activities:

1. Domain Analysis

- Select a real-world domain (e.g., hospital management, university course registration, e-commerce inventory) that the team members are commonly interested in.
- Describe the business context and operational goals
- Identify key entities (e.g., patients, courses, products)

2. Stakeholder Identification

- List primary users (e.g., administrators, customers, students)
- Define their roles and data access needs (e.g., query, maintain, report)

3. Requirement Gathering

- Conduct interviews, surveys, or literature reviews
- Document functional requirements (e.g., "Users can register for courses")
- Document non-functional requirements (e.g., "System must support 100 concurrent users")

4. Use Case Definition

- Create use case diagrams or narratives
- Describe typical user interactions with the system

5. Data Requirement Specification

- Identify required data fields and types (e.g., "each student has names, address, and date-of-birth")
- Define relationships between entities (e.g., "students enroll courses")
- Note constraints (e.g., "Each student can enroll in up to 5 courses")

6. Preliminary Planning

- Estimate data volume and growth
- Identify performance, security, and availability expectations

Deliverables

A **document** in portable document format (PDF) that contains the following information:

1. Proposed **project title** and **group ID**.
2. Group members that include **student names** and **IDs**.
3. **Declaration of AI tool usage**. (e.g., level 1 no AI, level 2 AI planning, level 3 AI as a service, level 4 AI collaboration and level 5 full AI) In this project, only level 1 and 2 are allowed.
4. **Business scenario** and **problem statement**: Briefly describe (**100 – 150 words**) the business context and operational goals that reflect the main problems that the selected database application tries to achieve. (e.g., no computerized system supported or with a partial computerized system supported or the current database application cannot fulfill the new requires for some specific reasons which you have to clearly specify).
5. **As-Is Analysis**: Briefly describe what are the key characteristics of the current situation (AS-IS) that include the following:
 - a. **Users**: List **at least two** groups of users (e.g., administrators, customers, students)
 - b. **Core entities** (**at least 6**): List the primary objects you intend to track (e.g., patients, courses, products, transactions).
 - c. **Key business rules**: List **at least three** specific rules your database will enforce (e.g., “A student cannot enroll in a course unless they have completed the prerequisite,” or “An order cannot be processed if stock is zero”)
6. **Proposed Database Application** (**no more than 6 pages**) which includes the following items:
 - a. **Mission statement** which is **a sentence** describing the goal of the database application that can supports all types of target users.
 - b. **Mission objectives** which describes particular tasks that the database must support. The mission objectives must conform to the mission statement.
 - c. **Scope and boundary** which identifies the key features of the database application with respect to the mission statement and objectives. The features may be presented in form of tasks, processes or data. Those features that are excluded from your database application can also be identified. The connection or relationship between what are included in and what are excluded from your database application can help clearly describe the scope and boundary of your database application. The scope and boundary can be shown as a Figure with one or more paragraphs describing the Figure.
 - d. **Mappings of major user views and operations** which define user roles and data access needs (e.g., query, maintain, report). It can be shown as table describing relationship among different groups of target users, different groups of data items, and different groups of operations.
 - e. **Users’ Requirements Specification** consists of (a) **Data Requirements** which describe the characteristics of each group of data items (**at least six data items**) used in your database application and (b) **Transaction Requirements** which describes how the data used in your database application (**at least six transactions**).
 - f. **Systems Requirements Specification** describes at least the following items: initial database size, database growth, the types and average number of record searches, networking and shared access requirements, performance, security, backup and recovery and legal issues.

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ITCS442 Relational Database Design

Submissions

Name the report file as <ProjectID>_<GID>_<last-three-digit-of-StudentID for every team member>_report.pdf, for example, "***P1_G01_999_988_977_report.pdf***".

Concerned Issues

1. Plagiarism is NOT allowed for any cases.
2. Creativity in design and presentation is preferable.

Evaluation

90% for project evaluation based on evaluation rubrics:

- instructor reviews: 70%
- peer reviews: 30%

10% for participation in peer reviews

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Evaluation Rubric

Criteria	Exemplary (3)	Proficient (2)	Developing (1)
Identifiers: Project Title, Group ID, Student Names and IDs and	All identifiers present.		Missing one or more identifiers
AI declaration	AI declaration clearly states level 1 or 2.	AI declaration is vague but present.	Missing AI declaration or level > 2.
Business scenario & Problem statement. (100 – 150 words)	Clearly identifies current pain points and why a new database application is needed.	Scenario is described but lacks specific operational goals.	Too short (< 100 words) or too long (> 150 words) to explain the problem clearly.
User groups	At least two user groups.		No or only one user group.
Core entities	At least six core entities.	Missing one or two core entities. (Note: at least six entities may be identified but some entities are not <i>core</i> entities)	Missing more than two core entities. (Note: at least six entities may be identified but many entities are not <i>core</i> entities)
Key business rules	At least three rules that are specific to the business.	Missing one business rules or many rules are generic (e.g., “save data”)	Missing multiple business rules or most rules are generic (e.g., “save data”)
Mission statement and objectives.	Mission statement is clearly identified and mission objectives are specific and aligned with the mission statement.	Mission statement is clearly identified but mission objectives are not specific or not aligned with mission statement.	Mission statement and objectives are missing or unclear.
Scopes and Boundary.	Clearly defines the scopes and boundary of the project, demonstrating a comprehensive understanding.	Provides a basic description of scopes and boundary, but lacks depth and clarity.	Scopes and boundary are not defined or are inaccurately described.
Mappings of Major User Views and Operations.	Provides comprehensive mappings of major user views and operations, considering user needs and interactions.		Incomplete or inaccurate mappings of user views and operations.
Data Requirements	The key characteristics of at least six data items covered all core entities are clearly identified.	The key characteristics of data items covered some core entities are clearly identified.	No or inadequate characteristics of data items are identified.
Transaction Requirements	At least six transactions are identified and they related to the key business rules .	At least six transactions are identified.	Less than six transactions are identified.
System Requirements	Clearly identifies and organizes with relevant justifications		Inadequate identification of requirements.